



MATH140: Limits and Continuity

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Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Limits by factoring

1. Evaluate: limit as x approaches 1 of $[1(x^2 - 1^2) / (x - 1)]$. (0/0 indeterminate form — factor to resolve.)

$$\lim_{x \rightarrow 1} \frac{1(x^2 - 1^2)}{x - 1}$$

Answer: _____

2. Evaluate: limit as x approaches 4 of $[1(x^2 - 4^2) / (x - 4)]$. (0/0 indeterminate form — factor to resolve.)

$$\lim_{x \rightarrow 4} \frac{1(x^2 - 4^2)}{x - 4}$$

Answer: _____

3. Evaluate: limit as x approaches 3 of $[4(x^2 - 3^2) / (x - 3)]$. (0/0 indeterminate form — factor to resolve.)

$$\lim_{x \rightarrow 3} \frac{4(x^2 - 3^2)}{x - 3}$$

Answer: _____

4. Evaluate: limit as x approaches 4 of $[2(x^2 - 4^2) / (x - 4)]$. (0/0 indeterminate form — factor to resolve.)

$$\lim_{x \rightarrow 4} \frac{2(x^2 - 4^2)}{x - 4}$$

Answer: _____



5. Evaluate: limit as x approaches 2 of $[5(x^2 - 2^2) / (x - 2)]$. (0/0 indeterminate form — factor to resolve.)

$$\lim_{x \rightarrow 2} \frac{5(x^2 - 2^2)}{x - 2}$$

Answer: _____

6. Evaluate: limit as x approaches 2 of $[5(x^2 - 2^2) / (x - 2)]$. (0/0 indeterminate form — factor to resolve.)

$$\lim_{x \rightarrow 2} \frac{5(x^2 - 2^2)}{x - 2}$$

Answer: _____

7. Evaluate: limit as x approaches 3 of $[1(x^2 - 3^2) / (x - 3)]$. (0/0 indeterminate form — factor to resolve.)

$$\lim_{x \rightarrow 3} \frac{1(x^2 - 3^2)}{x - 3}$$

Answer: _____

8. Evaluate: limit as x approaches 3 of $[1(x^2 - 3^2) / (x - 3)]$. (0/0 indeterminate form — factor to resolve.)

$$\lim_{x \rightarrow 3} \frac{1(x^2 - 3^2)}{x - 3}$$

Answer: _____

Limits by substitution

9. Evaluate the limit: limit as x approaches 1 of $(1x^2 + -1x + -4)$.

$$\lim_{x \rightarrow 1} (1x^2 - 1x - 4)$$

Answer: _____

10. The demand function for a product is $D(p) = -1p^2 + 11p + 99$. Find the limit of demand as price approaches \$1.

$$\lim_{x \rightarrow 1} (-1x^2 + 11x + 99)$$

Answer: _____



11. Evaluate: limit as x approaches -1 of $(3x^2 + 0x + 5)$.

$$\lim_{x \rightarrow -1} (3x^2 + 0x + 5)$$

Answer: _____

12. Evaluate the limit: limit as x approaches 2 of $(3x^2 + -4x + 5)$.

$$\lim_{x \rightarrow 2} (3x^2 - 4x + 5)$$

Answer: _____

13. The demand function for a product is $D(p) = -1p^2 + 5p + 101$. Find the limit of demand as price approaches \$3.

$$\lim_{x \rightarrow 3} (-1x^2 + 5x + 101)$$

Answer: _____

14. Evaluate: limit as x approaches -3 of $(1x^2 + -1x + 3)$.

$$\lim_{x \rightarrow -3} (1x^2 - 1x + 3)$$

Answer: _____

15. Evaluate the limit: limit as x approaches -1 of $(2x^2 + 3x + -3)$.

$$\lim_{x \rightarrow -1} (2x^2 + 3x - 3)$$

Answer: _____

16. The demand function for a product is $D(p) = -1p^2 + 3p + 95$. Find the limit of demand as price approaches \$4.

$$\lim_{x \rightarrow 4} (-1x^2 + 3x + 95)$$

Answer: _____

17. Evaluate: limit as x approaches 1 of $(5x^2 + -3x + 4)$.

$$\lim_{x \rightarrow 1} (5x^2 - 3x + 4)$$

Answer: _____

18. Evaluate the limit: limit as x approaches 3 of $(1x^2 + -2x + -4)$.

$$\lim_{x \rightarrow 3} (1x^2 - 2x - 4)$$

Answer: _____



19. The demand function for a product is $D(p) = -2p^2 + 10p + 113$. Find the limit of demand as price approaches \$3.

$$\lim_{x \rightarrow 3} (-2x^2 + 10x + 113)$$

Answer: _____

20. Evaluate: limit as x approaches 1 of $(1x^2 + -1x + 8)$.

$$\lim_{x \rightarrow 1} (1x^2 - 1x + 8)$$

Answer: _____

21. Evaluate the limit: limit as x approaches -2 of $(2x^2 + -4x + 6)$.

$$\lim_{x \rightarrow -2} (2x^2 - 4x + 6)$$

Answer: _____

22. The demand function for a product is $D(p) = -1p^2 + 7p + 103$. Find the limit of demand as price approaches \$2.

$$\lim_{x \rightarrow 2} (-1x^2 + 7x + 103)$$

Answer: _____

23. Evaluate: limit as x approaches -2 of $(4x^2 + -4x + 7)$.

$$\lim_{x \rightarrow -2} (4x^2 - 4x + 7)$$

Answer: _____

24. Evaluate the limit: limit as x approaches 0 of $(4x^2 + 2x + 1)$.

$$\lim_{x \rightarrow 0} (4x^2 + 2x + 1)$$

Answer: _____

25. The demand function for a product is $D(p) = -1p^2 + 4p + 110$. Find the limit of demand as price approaches \$4.

$$\lim_{x \rightarrow 4} (-1x^2 + 4x + 110)$$

Answer: _____



26. Evaluate: limit as x approaches 1 of $(4x^2 + -2x + -1)$.

$$\lim_{x \rightarrow 1} (4x^2 - 2x - 1)$$

Answer: _____

27. Evaluate the limit: limit as x approaches 1 of $(1x^2 + -4x + 5)$.

$$\lim_{x \rightarrow 1} (1x^2 - 4x + 5)$$

Answer: _____

28. The demand function for a product is $D(p) = -1p^2 + 11p + 44$. Find the limit of demand as price approaches \$2.

$$\lim_{x \rightarrow 2} (-1x^2 + 11x + 44)$$

Answer: _____

29. Evaluate: limit as x approaches -3 of $(1x^2 + -5x + -5)$.

$$\lim_{x \rightarrow -3} (1x^2 - 5x - 5)$$

Answer: _____

30. Evaluate the limit: limit as x approaches 0 of $(3x^2 + 3x + -8)$.

$$\lim_{x \rightarrow 0} (3x^2 + 3x - 8)$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Limits by factoring, Limits by substitution. All answers verified by independent computation.

Solutions



Limits by factoring

1. Evaluate: limit as x approaches 1 of $[1(x^2 - 1^2) / (x - 1)]$. (0/0 indeterminate form — factor to resolve.)

$$\lim_{x \rightarrow 1} \frac{1(x^2 - 1^2)}{x - 1}$$

→ Factor: $1(x-1)(x+1) / (x-1)$.

→ Cancel $(x-1)$ and substitute $x = 1$: $1(2 \cdot 1) = 2$.

Answer: $= (1)(x + 1)|_{x=1} = 2$

2. Evaluate: limit as x approaches 4 of $[1(x^2 - 4^2) / (x - 4)]$. (0/0 indeterminate form — factor to resolve.)

$$\lim_{x \rightarrow 4} \frac{1(x^2 - 4^2)}{x - 4}$$

→ Factor: $1(x-4)(x+4) / (x-4)$.

→ Cancel $(x-4)$ and substitute $x = 4$: $1(2 \cdot 4) = 8$.

Answer: $= (1)(x + 4)|_{x=4} = 8$

3. Evaluate: limit as x approaches 3 of $[4(x^2 - 3^2) / (x - 3)]$. (0/0 indeterminate form — factor to resolve.)

$$\lim_{x \rightarrow 3} \frac{4(x^2 - 3^2)}{x - 3}$$

→ Factor: $4(x-3)(x+3) / (x-3)$.

→ Cancel $(x-3)$ and substitute $x = 3$: $4(2 \cdot 3) = 24$.

Answer: $= (4)(x + 3)|_{x=3} = 24$

4. Evaluate: limit as x approaches 4 of $[2(x^2 - 4^2) / (x - 4)]$. (0/0 indeterminate form — factor to resolve.)

$$\lim_{x \rightarrow 4} \frac{2(x^2 - 4^2)}{x - 4}$$

→ Factor: $2(x-4)(x+4) / (x-4)$.

→ Cancel $(x-4)$ and substitute $x = 4$: $2(2 \cdot 4) = 16$.

Answer: $= (2)(x + 4)|_{x=4} = 16$

5. Evaluate: limit as x approaches 2 of $[5(x^2 - 2^2) / (x - 2)]$. (0/0 indeterminate form — factor to resolve.)

$$\lim_{x \rightarrow 2} \frac{5(x^2 - 2^2)}{x - 2}$$

→ Factor: $5(x-2)(x+2) / (x-2)$.

→ Cancel $(x-2)$ and substitute $x = 2$: $5(2 \cdot 2) = 20$.

Answer: $= (5)(x + 2)|_{x=2} = 20$



6. Evaluate: limit as x approaches 2 of $[5(x^2 - 2^2) / (x - 2)]$. (0/0 indeterminate form — factor to resolve.)

$$\lim_{x \rightarrow 2} \frac{5(x^2 - 2^2)}{x - 2}$$

→ Factor: $5(x-2)(x+2) / (x-2)$.

→ Cancel $(x-2)$ and substitute $x = 2$: $5(2 \cdot 2) = 20$.

Answer: $= (5)(x + 2)|_{x=2} = 20$

7. Evaluate: limit as x approaches 3 of $[1(x^2 - 3^2) / (x - 3)]$. (0/0 indeterminate form — factor to resolve.)

$$\lim_{x \rightarrow 3} \frac{1(x^2 - 3^2)}{x - 3}$$

→ Factor: $1(x-3)(x+3) / (x-3)$.

→ Cancel $(x-3)$ and substitute $x = 3$: $1(2 \cdot 3) = 6$.

Answer: $= (1)(x + 3)|_{x=3} = 6$

8. Evaluate: limit as x approaches 3 of $[1(x^2 - 3^2) / (x - 3)]$. (0/0 indeterminate form — factor to resolve.)

$$\lim_{x \rightarrow 3} \frac{1(x^2 - 3^2)}{x - 3}$$

→ Factor: $1(x-3)(x+3) / (x-3)$.

→ Cancel $(x-3)$ and substitute $x = 3$: $1(2 \cdot 3) = 6$.

Answer: $= (1)(x + 3)|_{x=3} = 6$



Limits by substitution

9. Evaluate the limit: limit as x approaches 1 of $(1x^2 + -1x + -4)$.

$$\lim_{x \rightarrow 1} (1x^2 - 1x - 4)$$

→ Polynomial — direct substitution: plug in $x = 1$.

$$\rightarrow = 1(1)^2 + -1(1) + -4 = -4.$$

Answer: $\lim_{x \rightarrow 1} = 1 + -1 + -4 = -4$

10. The demand function for a product is $D(p) = -1p^2 + 11p + 99$. Find the limit of demand as price approaches \$1.

$$\lim_{x \rightarrow 1} (-1x^2 + 11x + 99)$$

→ Substitute $p = 1$: $-1(1)^2 + 11(1) + 99 = 109$.

→ When price is \$1, demand approaches 109 units.

Answer: $\lim_{x \rightarrow 1} = -1 + 11 + 99 = 109$

11. Evaluate: limit as x approaches -1 of $(3x^2 + 0x + 5)$.

$$\lim_{x \rightarrow -1} (3x^2 + 0x + 5)$$

→ Direct substitution: $x = -1$.

$$\rightarrow = 3(-1)^2 + 0(-1) + 5 = 8.$$

Answer: $\lim_{x \rightarrow -1} = 3 + 0 + 5 = 8$

12. Evaluate the limit: limit as x approaches 2 of $(3x^2 + -4x + 5)$.

$$\lim_{x \rightarrow 2} (3x^2 - 4x + 5)$$

→ Polynomial — direct substitution: plug in $x = 2$.

$$\rightarrow = 3(2)^2 + -4(2) + 5 = 9.$$

Answer: $\lim_{x \rightarrow 2} = 12 + -8 + 5 = 9$

13. The demand function for a product is $D(p) = -1p^2 + 5p + 101$. Find the limit of demand as price approaches \$3.

$$\lim_{x \rightarrow 3} (-1x^2 + 5x + 101)$$

→ Substitute $p = 3$: $-1(3)^2 + 5(3) + 101 = 107$.

→ When price is \$3, demand approaches 107 units.

Answer: $\lim_{x \rightarrow 3} = -9 + 15 + 101 = 107$



14. Evaluate: limit as x approaches -3 of $(1x^2 + -1x + 3)$.

$$\lim_{x \rightarrow -3} (1x^2 - 1x + 3)$$

→ Direct substitution: $x = -3$.

$$\rightarrow = 1(-3)^2 + -1(-3) + 3 = 15.$$

Answer: $\lim_{x \rightarrow -3} = 9 + 3 + 3 = 15$

15. Evaluate the limit: limit as x approaches -1 of $(2x^2 + 3x + -3)$.

$$\lim_{x \rightarrow -1} (2x^2 + 3x - 3)$$

→ Polynomial — direct substitution: plug in $x = -1$.

$$\rightarrow = 2(-1)^2 + 3(-1) + -3 = -4.$$

Answer: $\lim_{x \rightarrow -1} = 2 + -3 + -3 = -4$

16. The demand function for a product is $D(p) = -1p^2 + 3p + 95$. Find the limit of demand as price approaches \$4.

$$\lim_{x \rightarrow 4} (-1x^2 + 3x + 95)$$

→ Substitute $p = 4$: $-1(4)^2 + 3(4) + 95 = 91$.

→ When price is \$4, demand approaches 91 units.

Answer: $\lim_{x \rightarrow 4} = -16 + 12 + 95 = 91$

17. Evaluate: limit as x approaches 1 of $(5x^2 + -3x + 4)$.

$$\lim_{x \rightarrow 1} (5x^2 - 3x + 4)$$

→ Direct substitution: $x = 1$.

$$\rightarrow = 5(1)^2 + -3(1) + 4 = 6.$$

Answer: $\lim_{x \rightarrow 1} = 5 + -3 + 4 = 6$

18. Evaluate the limit: limit as x approaches 3 of $(1x^2 + -2x + -4)$.

$$\lim_{x \rightarrow 3} (1x^2 - 2x - 4)$$

→ Polynomial — direct substitution: plug in $x = 3$.

$$\rightarrow = 1(3)^2 + -2(3) + -4 = -1.$$

Answer: $\lim_{x \rightarrow 3} = 9 + -6 + -4 = -1$

19. The demand function for a product is $D(p) = -2p^2 + 10p + 113$. Find the limit of demand as price approaches \$3.

$$\lim_{x \rightarrow 3} (-2x^2 + 10x + 113)$$

→ Substitute $p = 3$: $-2(3)^2 + 10(3) + 113 = 125$.

→ When price is \$3, demand approaches 125 units.

Answer: $\lim_{x \rightarrow 3} = -18 + 30 + 113 = 125$



20. Evaluate: limit as x approaches 1 of $(1x^2 + -1x + 8)$.

$$\lim_{x \rightarrow 1} (1x^2 - 1x + 8)$$

→ Direct substitution: $x = 1$.

$$\rightarrow = 1(1)^2 + -1(1) + 8 = 8.$$

Answer: $\lim_{x \rightarrow 1} = 1 + -1 + 8 = 8$

21. Evaluate the limit: limit as x approaches -2 of $(2x^2 + -4x + 6)$.

$$\lim_{x \rightarrow -2} (2x^2 - 4x + 6)$$

→ Polynomial — direct substitution: plug in $x = -2$.

$$\rightarrow = 2(-2)^2 + -4(-2) + 6 = 22.$$

Answer: $\lim_{x \rightarrow -2} = 8 + 8 + 6 = 22$

22. The demand function for a product is $D(p) = -1p^2 + 7p + 103$. Find the limit of demand as price approaches \$2.

$$\lim_{x \rightarrow 2} (-1x^2 + 7x + 103)$$

→ Substitute $p = 2$: $-1(2)^2 + 7(2) + 103 = 113$.

→ When price is \$2, demand approaches 113 units.

Answer: $\lim_{x \rightarrow 2} = -4 + 14 + 103 = 113$

23. Evaluate: limit as x approaches -2 of $(4x^2 + -4x + 7)$.

$$\lim_{x \rightarrow -2} (4x^2 - 4x + 7)$$

→ Direct substitution: $x = -2$.

$$\rightarrow = 4(-2)^2 + -4(-2) + 7 = 31.$$

Answer: $\lim_{x \rightarrow -2} = 16 + 8 + 7 = 31$

24. Evaluate the limit: limit as x approaches 0 of $(4x^2 + 2x + 1)$.

$$\lim_{x \rightarrow 0} (4x^2 + 2x + 1)$$

→ Polynomial — direct substitution: plug in $x = 0$.

$$\rightarrow = 4(0)^2 + 2(0) + 1 = 1.$$

Answer: $\lim_{x \rightarrow 0} = 0 + 0 + 1 = 1$



25. The demand function for a product is $D(p) = -1p^2 + 4p + 110$. Find the limit of demand as price approaches \$4.

$$\lim_{x \rightarrow 4} (-1x^2 + 4x + 110)$$

→ Substitute $p = 4$: $-1(4)^2 + 4(4) + 110 = 110$.

→ When price is \$4, demand approaches 110 units.

Answer: $\lim_{x \rightarrow 4} = -16 + 16 + 110 = 110$

26. Evaluate: limit as x approaches 1 of $(4x^2 + -2x + -1)$.

$$\lim_{x \rightarrow 1} (4x^2 - 2x - 1)$$

→ Direct substitution: $x = 1$.

→ $= 4(1)^2 + -2(1) + -1 = 1$.

Answer: $\lim_{x \rightarrow 1} = 4 + -2 + -1 = 1$

27. Evaluate the limit: limit as x approaches 1 of $(1x^2 + -4x + 5)$.

$$\lim_{x \rightarrow 1} (1x^2 - 4x + 5)$$

→ Polynomial — direct substitution: plug in $x = 1$.

→ $= 1(1)^2 + -4(1) + 5 = 2$.

Answer: $\lim_{x \rightarrow 1} = 1 + -4 + 5 = 2$

28. The demand function for a product is $D(p) = -1p^2 + 11p + 44$. Find the limit of demand as price approaches \$2.

$$\lim_{x \rightarrow 2} (-1x^2 + 11x + 44)$$

→ Substitute $p = 2$: $-1(2)^2 + 11(2) + 44 = 62$.

→ When price is \$2, demand approaches 62 units.

Answer: $\lim_{x \rightarrow 2} = -4 + 22 + 44 = 62$

29. Evaluate: limit as x approaches -3 of $(1x^2 + -5x + -5)$.

$$\lim_{x \rightarrow -3} (1x^2 - 5x - 5)$$

→ Direct substitution: $x = -3$.

→ $= 1(-3)^2 + -5(-3) + -5 = 19$.

Answer: $\lim_{x \rightarrow -3} = 9 + 15 + -5 = 19$



30. Evaluate the limit: limit as x approaches 0 of $(3x^2 + 3x - 8)$.

$$\lim_{x \rightarrow 0} (3x^2 + 3x - 8)$$

→ Polynomial — direct substitution: plug in $x = 0$.

$$\rightarrow = 3(0)^2 + 3(0) - 8 = -8.$$

Answer: $\lim_{x \rightarrow 0} = 0 + 0 - 8 = -8$

